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Aryan Kumar Jha

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Passport: X4222777 **Gender:** Male **Date of birth:** 09/01/2007 **Place of birth:** Jaipur, India **Nationality:** Indian

EDUCATION AND TRAINING

[2024 – Current] **Bachelor of Technology - Energy Sciences and Engineering**

IIT Delhi - Abu Dhabi https://iitdabudhabi.ac.ae/

City: Abu Dhabi | **Country:** United Arab Emirates |

Understanding energy systems and how material sciences, computational intelligent systems, policies and economics can be harnessed to accelerate the energy transition.

[2022 – 2024] **12th Grade**

Hansraj School

City: Delhi | **Country:** India | **Field(s) of study:** Natural sciences, mathematics and statistics: ● *Physics* ● *Mathematics* ● *Chemistry ;* Information and Communication Technologies: ● *Software and applications development and analysis*

[2016 – 2022] **10th Matriculation**

Teoler High School

City: Jaipur | **Country:** India | **Field(s) of study:** Natural sciences, mathematics and statistics: ● *Mathematics* ● *Physics* ● *Chemistry ;* Arts and humanities : ● *Literature and linguistics ;* Information and Communication Technologies: ● *Software and applications development and analysis*

WORK EXPERIENCE

MISN Lab - IIT Delhi

City: Delhi | **Country:** India

[2025 – Current] **Research Intern**

Machine Intelligence Signal and Network Lab is a research group of IIT Delhi's School of AI led by Prof. Sandeep Kumar specialising in applying AI to various challenges faced by the Indian demographic.

As of now, we are being funded by the union govt. for Project Adivaani - an all-inclusive NMT system for Indian tribal languages

My involvement till now is

- using open-source LLMs such as gemma2-7B for zero-shot learning for cross-lingual translation
- organizing and creating datasets for Indian low resource tribal languages
- Supervised fine-tuning using HPC, open-source models and back-translation techniques
- contributing to research papers on low-resource ML systems

CERTIFICATIONS

[One Fourth Labs - IIT Madras, 2022]

PadhAI - Deep Learning Certification

One Fourth Labs is an incubent startup of IIT Madras, tackling challenges of the Indian demographic using data science, machine learning and deep learning techniques. They organised a course on Deep Learning where I learnt

- information theory
- intermediate data manipulation and processing
- feed-forward neural networks
- CNN's, RNN's, Ensemble models, Encoder-Decoder architecture
- application of DL in tasks like translation, computer vision, NLP, object detection

All these tools were ultimately used in the AI4Bharat initiative.

Mode of learning: Online

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING B1 **READING** B1 **WRITING** B1

SPOKEN PRODUCTION B1 **SPOKEN INTERACTION** B1

German

LISTENING A1 **READING** A1 **WRITING** A1

SPOKEN PRODUCTION A1 **SPOKEN INTERACTION** A1

PROJECTS

[2025 – Current] **Subsurface Characterisation geothermal hot spring potential estimation**

Attempting to replicate the subsurface characterisation models built by NREL using MLP, CNN and reinforcement learning techniques and publicly available data sources to understand real-world application of AI systems in deep science technologies. The paper published by the team is available at -

https://gdr.openei.org/submissions/1346

Link: https://github.com/AryanSkywalker/geothermal_osr_replication

[2025] **Bhilli to Hindi LLM Prompting Study**

Analysed the performance of zero-shot learning for various open - source LLM's such as Gemma-2-9B, Mistral-7B, Bloomz-7B1, Qwen3-8B and more on NMT related-objectives

Link: https://github.com/AryanSkywalker/bhili-hindi_prompting_study

[2020] **Hindi Character Detection and Recognition**

- Training and deploying deep learning techniques to detect and recognize hindi characters from images of signboards
- Used PyTorch, OpenCV2 to train dense RCNN, CNN and LSTM for object detection, recognition and subsequent transliteration to the English language

Link: https://github.com/AryanSkywalker/AS_P2_Hindi-Character-Detection

[2021] **Hand Gesture Recognition**

- Training convolution neural techniques to recognize and interpret hand signs from real-time camera feed
- Collecting hand-made data from camera feed for real-time calibration and augmentation

Link: https://github.com/AryanSkywalker/AS-P1-Gesture-Recognition

[2022] **Computer vision based answer sheet checker**

Using image processing and statistical modelling techniques to process, segment and check multi-choice answer sheets

Created a simple app that underfunded schools can use to grade papers efficiently

Link: https://github.com/AryanSkywalker/AS_P4_OMR_Mimic

NETWORKS AND MEMBERSHIPS

[11/2025 – Current] **ACS Student Chapter, IIT Delhi - Abu Dhabi** Abu Dhabi

Organised ACS Ignite, a flagship networking event, attended by representatives of educational institutions, independent startups and researches to deliver talks, seminars and encourage collaboration on the applications of deep science to world challenges

Link: https://iitdabudhabi.ac.ae/acsignite

HONOURS AND AWARDS

[2024] **Qualified JEE Advance**

[2023] **Qualified Zonal Computing Olympiad, New Delhi Zone**

[2021] **Qualified Zonal Informatics Olympiad, Jaipur Zone**

[2022] **Silver Medal, Rank 2 - STSE Rajasthan**

VOLUNTEERING

[2025] **Oorja Sangam - National Workshop, IIT Delhi** Delhi

Organisation and event management for seminars and talks held during the workshop, attended by academics from all around India and the education minister - Shri Dharmendra Pradhan on the topic encompassing "Energy Education in India"

Link: https://oorjasangam.iitd.ac.in/